

RaceDay API Endpoint Plan

Part 1 – System Planning and Database

Roles

- **Organiser:** creates and manages events and categories, views enrolments, and records/updates results.
- **Participant:** manages their profile, enrolls in available categories, and views their enrolments and results.
- **Public:** can register/login and view public event, category and result information.

Planned API Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new Organiser or Participant account.	Public	{ firstName, lastName, email, password, role, phone }	201 Created; 400 Bad Request; 409 Conflict.
POST	/api/auth/login	Authenticates a user and returns an access token.	Public	{ email, password }	200 OK; 401 Unauthorized.
GET	/api/users/me	Returns the profile of the logged-in user.	Any logged-in user	None	200 OK; 401 Unauthorized.
PUT	/api/users/me	Updates the logged-in user's profile.	Any logged-in user	{ firstName, lastName, phone }	200 OK; 400 Bad Request; 401 Unauthorized.
GET	/api/events	Returns available RaceDay events.	Public	None	200 OK - event list.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}	Returns one event with its venue and categories.	Public	None	200 OK; 404 Not Found.
POST	/api/events	Creates a new race event.	Organiser	{ venueId, eventName, description, eventDate, startTime, status }	201 Created; 400 Bad Request; 403 Forbidden.
PUT	/api/events/{id}	Updates an event managed by the organiser.	Organiser	{ venueId, eventName, description, eventDate, startTime, status }	200 OK; 404 Not Found; 403 Forbidden.
DELETE	/api/events/{id}	Removes an event where deletion is permitted.	Organiser	None	204 No Content; 404 Not Found; 409 Conflict.
GET	/api/categories	Returns race categories, optionally filtered by event.	Public	None	200 OK - category list.
GET	/api/events/{eventId}/categories	Returns categories belonging to an event.	Public	None	200 OK; 404 Not Found.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/categories/{categoryId}	Creates a category for an event.	Organiser	{ category-Name, distanceKm, maxParticipants, entryFee }	201 Created; 400 Bad Request; 403 Forbidden.
PUT	/api/categories/{categoryId}	Updates a race category.	Organiser	{ category-Name, distanceKm, maxParticipants, entryFee }	200 OK; 404 Not Found; 403 Forbidden.
DELETE	/api/categories/{categoryId}	Deletes a category where permitted.	Organiser	None	204 No Content; 404 Not Found; 409 Conflict.
POST	/api/events/{eventId}/enrolments	Enrols a logged-in participant in a category.	Participant	{ categoryId }	201 Created; 400 Bad Request; 409 Conflict.
GET	/api/enrolments/{enrolmentId}	Returns the logged-in participant's enrolments.	Participant	None	200 OK; 401 Unauthorized.
GET	/api/events/{eventId}/enrolments	Returns enrolments for organiser administration.	Organiser	None	200 OK; 403 Forbidden; 404 Not Found.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/enrolments/{id}	Creates an enrolment status where permitted.	Organiser	{ status }	200 OK; 404 Not Found; 403 Forbidden.
DELETE	/api/enrolments/{id}	Deletes the logged-in participant's enrolment.	Participant	None	204 No Content; 404 Not Found; 409 Conflict.
POST	/api/results	Records a result for a completed enrolment.	Organiser	{ enrolmentId, finishPosition, finishTime, resultStatus }	201 Created; 400 Bad Request; 409 Conflict.
GET	/api/events/{eventId}/results	Returns event results ordered by finish position.	Public	None	200 OK; 404 Not Found.
GET	/api/results/{id}	Returns results for the logged-in participant.	Participant	None	200 OK; 401 Unauthorized.
PUT	/api/results/{id}	Corrects or updates a recorded result.	Organiser	{ finishPosition, finishTime, resultStatus }	200 OK; 404 Not Found; 403 Forbidden.
DELETE	/api/results/{id}	Removes an incorrect result.	Organiser	None	204 No Content; 404 Not Found; 403 Forbidden.

Planning Rules

- All endpoints use the `/api/` prefix.
- Authentication endpoints are public.
- Protected endpoints require authentication.
- Organiser and Participant permissions are enforced using role-based authorization.
- Part 2 implementation should closely follow these routes and response codes.